12 Lecture 12 (February 10, 2025, est.): From a fiber functor to a coalgebra

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Continue with the exact faithful k-linear functor $\omega : \mathcal{C} \to \mathrm{Vec}_k$ and the objects P_X and algebras A_X of Lecture 11.

The local equivalence

12.1 Theorem. *The functor $\widetilde{\omega}_X$ is an equivalence*

$$\langle X\rangle_{\oplus} \simeq A_X\text{-mod}_{\mathrm{fd}}, \qquad A_X \simeq \mathrm{End}(\omega|_{\langle X\rangle_{\oplus}}).$$

Proof. Write $A = A_X$, $P = P_X$, and $V = \omega X$. Precomposition by $a \in A$ defines a morphism on $H_X = V^* \otimes X$ whose image under ω sends b to ba. It preserves P, because $\omega P = A$ is closed under multiplication and ω reflects subobject inclusions. These maps make P a right A-module object in $\mathcal{C}$. Its image under ω is the right regular module. Its natural left A-action under $\widetilde{\omega}_X$ is left multiplication, so the two actions commute.

For a finite-dimensional left A-module M, form the object

$$Q(M) = P \otimes_A M = \mathrm{coker}\big(P \otimes A \otimes M \xrightarrow{r\otimes \mathrm{id} - \mathrm{id}\otimes \mu_M} P \otimes M\big).$$

The products with vector spaces are the external products of Lecture 11. This cokernel belongs to $\langle X\rangle_{\oplus}$, and exactness gives, naturally and as left A-modules,

$$\widetilde{\omega}_X Q(M) \simeq A \otimes_A M \simeq M.$$

To identify the other composite, start with the evaluation morphism

$$P \otimes \omega X \longrightarrow X,$$

obtained by restricting evaluation from $H_X \otimes V$. For $X^{\oplus n}$ take its direct sum. If $Y \subseteq X^{\oplus n}$, its restriction to $P \otimes \omega Y$ factors through Y: after applying ω, this says precisely that A preserves ωY. Exact faithfulness detects the required zero composite to the quotient. Passing to quotients defines evaluation for every subquotient $Y \in \langle X\rangle_{\oplus}$. The graph argument from the previous lecture makes these evaluations independent of the presentation and natural in Y.

They are balanced over A, so they induce

$$Q(\widetilde{\omega}_X Y) = P \otimes_A \omega Y \longrightarrow Y.$$

Applying ω gives the isomorphism $A \otimes_A \omega Y \to \omega Y$. An exact faithful functor reflects isomorphisms, by applying it to kernel and cokernel. Thus this evaluation is an isomorphism. The two natural isomorphisms exhibit Q as a quasi-inverse, proving the equivalence.

Finally, A acts naturally on all these vector spaces, giving a map to the endomorphism algebra of the restricted functor. Conversely, a natural endomorphism is determined by its component on X, since all objects are subquotients of finite sums of X. Naturality forces that component to preserve every $\omega Y \subseteq \omega(X^{\oplus n})$, so it lies in A. These constructions are inverse algebra maps. ∎

Passing from local algebras to one coalgebra

Set $B_X = A_X^*$. The preceding theorem and algebra–coalgebra duality give

$$\langle X\rangle_\oplus \simeq \mathrm{Comod}_{\mathrm{fd}}(B_X).$$

If $\langle X\rangle_\oplus \subseteq \langle Z\rangle_\oplus$, restriction of natural endomorphisms gives $A_Z \to A_X$. This map is surjective. Indeed, under the local equivalence for Z, put $M = \omega X$ and $D = A_Z/\operatorname{Ann}(M)$. The D-module M is faithful. Choosing a basis $m_1, \dots, m_d$ embeds the regular module by

$$D \longrightarrow M^{\oplus d}, \qquad a \longmapsto (am_1, \dots, am_d).$$

Every finite-dimensional D-module is a quotient of a finite direct sum of copies of D, hence a subquotient of copies of M. Thus the subcategory generated by M is exactly D-$\mathrm{mod}_{\mathrm{fd}}$, and $A_X \simeq D$. Dualizing gives an injection $B_X \hookrightarrow B_Z$.

Direct sums of objects make these subcategories a directed system. Define

$$B = \varinjlim_X B_X = \bigcup_X B_X$$

with the compatible coalgebra structures. Every object and every finite collection of morphisms of $\mathcal{C}$ belongs to some $\langle X\rangle_\oplus$. The local equivalences therefore assemble into a fully faithful functor $\mathcal{C} \to \mathrm{Comod}_{\mathrm{fd}}(B)$. It is essentially surjective: the finitely many matrix coefficients of a finite-dimensional B-comodule lie in some B_X, so its coaction already factors through that subcoalgebra. We have proved the abelian reconstruction theorem

$$\mathcal{C} \simeq \mathrm{Comod}_{\mathrm{fd}}(B), \tag{12.1}$$

compatibly with ω.

Natural endomorphisms and matrix coefficients

Let A now be any finite-dimensional algebra and let $U : A\text{-}\mathrm{mod}_{\mathrm{fd}} \to \mathrm{Vec}_k$ forget the action. For a commutative algebra R, put $U_R = U(-) \otimes R$. There is an algebra isomorphism

$$A \otimes R \simeq \mathrm{End}(U_R).$$

One direction lets $A \otimes R$ act. For the other, send a natural endomorphism λ to $\lambda_A(1 \otimes 1)$. For $m \in M$, the A-linear map $A \to M$, $a \mapsto am$, gives

$$\lambda_M(m \otimes 1) = \lambda_A(1 \otimes 1) \cdot m.$$

This proves that the two constructions are inverse. Notice that the components of λ need only be R-linear, not A-linear.

Applying this locally and passing to inverse limits yields

$$\mathrm{End}(\omega_R) \simeq \varprojlim_X (A_X \otimes R) \simeq \mathrm{Hom}_k(B, R).$$

The multiplication on the right is convolution using the coproduct of B.

It is useful to describe B without the directed system. For $v \in \omega X$ and $\phi \in (\omega X)^*$, write $[\phi \otimes v]_X$ for a matrix coefficient. Then

$$B \simeq \left(\bigoplus_X (\omega X)^* \otimes \omega X \right) / \mathcal{R},$$

where a small skeleton is used and $\mathcal{R}$ is spanned by

$$[\phi \otimes \omega(f)v]_Y - [\phi \circ \omega(f) \otimes v]_X, \qquad f : X \to Y.$$

Indeed, a linear map from this quotient to any vector space encodes precisely a natural family of linear maps on the spaces ωX with values in their tensor products with that vector space. On each local subcategory the preceding calculation identifies the quotient with A_X^*; taking the directed union gives B. If e_i is a basis of ωX with dual basis e_i^*, the coalgebra structure and universal coaction are

$$\begin{aligned} \Delta[\phi \otimes v]_X &= \sum_i [\phi \otimes e_i]_X \otimes [e_i^* \otimes v]_X, \\ \varepsilon[\phi \otimes v]_X &= \phi(v), \\ \rho_X(v) &= \sum_i e_i \otimes [e_i^* \otimes v]_X. \end{aligned}$$

The expressions are independent of the basis by the canonical tensor representing the identity.

In particular, applying this construction to the forgetful functor on finite-dimensional comodules of a coalgebra C recovers C: its finite-dimensional coefficient subcoalgebras exhaust it. A coalgebra map $C \to C'$ gives a functor on comodules by composing coactions. Conversely, a functor compatible with the forgetful functors sends the coefficient $[\phi \otimes v]_X$ to $[\phi \otimes v]_{F(X)}$; naturality makes this well-defined and the formulas above show it is a coalgebra map. This is the functorial form of abelian reconstruction used next.

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